Lab 3:

Task: Disaster Relief Route Planner

Scenario:
After a flood, a relief convoy must deliver supplies from a depot to a shelter across a 7x7 city grid. Each cell has a terrain type with a different crossing cost: 1 = Clear Road, 3 = Debris, 6 = Flooded, 0 = Impassable. The convoy can move Up, Down, Left, or Right, and wants the route with the lowest total cost, not the fewest cells.

The Grid:

```
# 0 = Impassable, 1 = Clear Road, 3 = Debris, 6 = Flooded
grid = [
  [1, 1, 1, 0, 1, 1, 1],
  [1, 0, 1, 0, 1, 0, 1],
  [1, 0, 3, 3, 3, 0, 1],
  [1, 0, 3, 6, 3, 0, 1],
  [1, 0, 3, 3, 3, 0, 1],
  [1, 0, 0, 0, 0, 0, 1],
  [1, 1, 1, 1, 1, 1, 1],
]

depot = (0, 0)
shelter = (6, 6)
```

Tasks:

1. State the search problem formulation (State, Initial State, Actions, Goal Test, Path Cost) for this grid, in a markdown cell.

2. Design an admissible heuristic for this grid and justify it in 2–3 sentences before writing any code — your justification must explicitly reference the cheapest possible terrain cost on your grid.
3. Implement get_neighbors(state, grid) from scratch.
4. Implement greedy_search(grid, start, goal) from scratch, using heapq, returning (path, total_cost, nodes_explored).
5. Implement a_star_search(grid, start, goal) from scratch, using heapq, returning (path, total_cost, nodes_explored).
6. Run both algorithms on your grid and print each one's path, total cost, and nodes explored.
7. In a markdown cell, answer the following in 5–7 sentences total:
    - Did Greedy and A* return different paths on your grid? If yes, explain exactly which cell(s) caused Greedy to make a worse choice, using your heuristic values. If no, redesign your grid until they *do* differ, then explain the change you made and why it worked.
    - Your convoy planner is told: "Flooded cells may become Impassable without warning as water rises." Without writing code, explain in 2–3 sentences whether this changes the environment classification from Lab 2 (static vs. dynamic, deterministic vs. stochastic), and what that would mean for how often the route needs to be recalculated.